Hamiltonian of phonons:

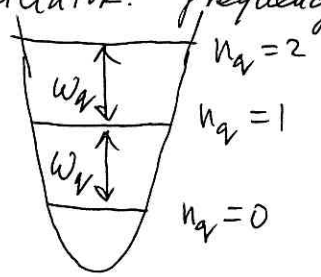
$$\hat{H}_{ph} = \sum_q \hbar\omega_q \left(b_q^+ b_q + \frac{1}{2} \right) =$$

$$= \sum_q \hbar\omega_q \left(\hat{n}_q + \frac{1}{2} \right) = \sum_q \hat{H}_{SHO}(q)$$

hamiltonian of q-th simple harmonic oscillator.

where $\hat{n}_q = b_q^+ b_q$ is the occupation number operator which counts number of phonons in q-th harmonic oscillator.

* q-th oscillator: frequency $\omega_q = 2\omega_0 \left| \sin \frac{qa}{2} \right|$.



Notice that number of phonons n_q with momentum q ~~is~~ the same as ~~the~~ specifying that the oscillator is in n_q-th excited state.

The energy of the state with $\underline{n_q}$ phonons is $\underline{\hbar\omega_q \left(n_q + \frac{1}{2} \right)}$.

Modes with different wavenumbers q are independent from each other $\Rightarrow$ partition function of the lattice

$$Z = \prod_q Z_q$$

is the product of partition functions describing oscillators with frequency ω_q.

Thus
$$Z_q = \sum_{n_q=0}^{\infty} e^{-\frac{\hbar\omega_q}{\tau}\left(n_q+\frac{1}{2}\right)} = \frac{e^{-\frac{\hbar\omega_q}{2\tau}}}{1-e^{-\frac{\hbar\omega_q}{\tau}}} = \frac{1}{2\sinh\left(\frac{\hbar\omega_q}{2\tau}\right)}$$

Number of phonons in q-th oscillator = occupation number of q-th oscillator

$$\langle n_q \rangle = \frac{\sum_{n_q=0}^{\infty} n_q\, e^{-\frac{\hbar\omega_q}{\tau}\left(n_q+\frac{1}{2}\right)}}{Z_q} =$$

$$= \frac{e^{-\frac{\hbar\omega_q}{2\tau}}}{Z_q} \sum_{n_q=0}^{\infty} n_q\, e^{-\frac{\hbar\omega_q}{\tau} n_q} = \frac{e^{-\frac{\hbar\omega_q}{2\tau}}}{Z_q}\left(-\frac{\partial}{\partial\left(\frac{\hbar\omega_q}{\tau}\right)} \sum_{n_q} e^{-\frac{\hbar\omega_q}{\tau} n_q}\right)$$

$$= \frac{e^{-\frac{\hbar\omega_q}{2\tau}}}{Z_q}\left(-\frac{\partial}{\partial\left(\frac{\hbar\omega_q}{\tau}\right)} \frac{1}{1-e^{-\frac{\hbar\omega_q}{\tau}}}\right)$$

But $\frac{\partial}{\partial x}\frac{1}{1-e^{-x}} = \frac{-e^{-x}}{(1-e^{-x})^2}$

$$\Rightarrow \langle n_q \rangle = \left(1-e^{-\frac{\hbar\omega_q}{\tau}}\right)\frac{e^{-\frac{\hbar\omega_q}{\tau}}}{\left(1-e^{-\frac{\hbar\omega_q}{\tau}}\right)^2} = \frac{e^{-\frac{\hbar\omega_q}{\tau}}}{1-e^{-\frac{\hbar\omega_q}{\tau}}} = \frac{1}{e^{\frac{\hbar\omega_q}{\tau}}-1}$$

$\boxed{\text{Planck distribution function}}$

Thermal energy of q-th oscillator

$$U_q = \langle \varepsilon_q \rangle = \hbar\omega_q \langle n_q \rangle = \frac{\hbar\omega_q}{e^{\frac{\hbar\omega_q}{\tau}} - 1}$$

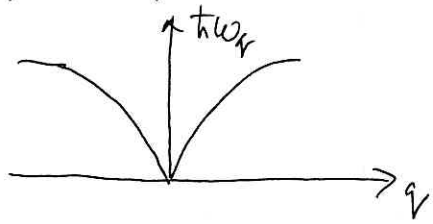
$\Rightarrow$ thermal energy of phonons gas is obtained by summing over all q from the Brillouin zone

$$U = \sum_q U_q = \sum_q \frac{\hbar\omega_q}{e^{\frac{\hbar\omega_q}{\tau}} - 1}$$

In our one-dimensional chain $\hbar\omega_q = 2\omega_0 \left|\sin\frac{qa}{2}\right|$ $\left(\omega_0 = \sqrt{\frac{K}{m}}\right)$ and $q \to q_n = \frac{2\pi n}{Na}$, with N independent q values.

$\boxed{D=3}$ In three-dimensional crystals $\hbar\omega_q$ has similar structure:

$\hbar\omega_q = vq$ at small $q = |\vec{q}| = \sqrt{q_x^2 + q_y^2 + q_z^2}$



and reaches maximum at the boundaries of 3-dimensional Brillouin zone.

At low temperature τ phonons with energies $\hbar\omega_q \sim \tau$ are most important $\Rightarrow$ we can approximate

$\omega_q = vq$, where $v \approx \sqrt{\frac{K}{m}}$ is the speed of sound.

Each atom has 3 independent directions to oscillate

$\Rightarrow$ there $3N$ independent oscillators in that case.

This is described by having N oscillators with 3 independent polarizations: $\begin{cases} 2 \text{ transverse} \\ 1 \text{ longitudinal.} \end{cases}$

$$N \cdot 3 = 3N.$$

Evaluate sum over (q) by converting it into integral. For example: consider sum over q_x

$$\sum_{q_x} f(q_x) = \sum_{n_x=-\frac{N}{2}}^{N/2} f\left(\frac{2\pi}{L_x} n_x\right) = \int_{-\frac{N}{2}}^{\frac{N}{2}} dn_x \, f\left(\frac{2\pi}{L_x} n_x\right)$$

$$q_x = \frac{2\pi n_x}{Na} = \frac{2\pi}{L_x} n_x$$

but $dn_x = \frac{L_x}{2\pi} dq_x \Rightarrow$

$$\int_{-N/2}^{N/2} dn_x \, f\left(\frac{2\pi n_x}{L_x}\right) = \frac{L_x}{2\pi} \int_{-\pi/a}^{\pi/a} dq_x \, f(q_x)$$

The factor $\frac{L_x}{2\pi} \equiv \frac{dq_x}{(\Delta q_x)} = \frac{dq_x}{(2\pi/L_x)}$ is the density of states for considered 1d example.

It counts number of modes in interval dq_x by dividing the interval by the minimal step $\Delta q_x = \frac{2\pi}{L_x}$.

$\boxed{D=3}$ Then $\sum_{\vec{q}} f(q) = \sum_{q_x, q_y, q_z} f(q_x, q_y, q_z) =$

$$= \underbrace{\frac{L_x}{2\pi} \frac{L_y}{2\pi} \frac{L_z}{2\pi}}_{\frac{V}{(2\pi)^3} = \text{3-dimensional density of states}} \int_{-\pi/a}^{\pi/a} dq_x \, dq_y \, dq_z \, f(q_x, q_y, q_z).$$

For spherically isotropic crystal $dq_x dq_y dq_z = 4\pi q^2 dq$

$$\Rightarrow \sum_q f(q) = \frac{V}{(2\pi)^3} 4\pi \int_0^{q_D} dq \, q^2 f(q).$$

(Everywhere above single polarization is assumed for simplicity!)

The upper limit of integration, q_d, is related to Debye frequency $\omega_d = v q_d$.
We determine it by requiring

$$\sum_q 1 = 3N = \text{total number of } q\text{-modes,}$$

including 3 polarizations.

Thus $$(3) \cdot \frac{4\pi V}{(2\pi)^3} \cdot \underbrace{\int_0^{q_d} dq\, q^2}_{\frac{1}{3} q_d^3} = 3N$$

$$\frac{V}{2\pi^2} q_d^3 = 3N \Rightarrow q_d = \left(\frac{6\pi^2 N}{V}\right)^{1/3}.$$

$\Rightarrow$ Debye frequency $\omega_d = v q_d$.

Then finally

$$U = \sum_q \frac{\hbar\omega_q}{e^{\frac{\hbar\omega_q}{\tau}} - 1} = (3) \frac{V}{2\pi^2} \int_0^{q_d} dq\, q^2 \frac{\hbar v q}{e^{\frac{\hbar v q}{\tau}} - 1}.$$

Introduce dimensionless $x = \frac{\hbar v q}{\tau} \Rightarrow dq = \frac{\tau}{\hbar v} dx$

$$U = \frac{3V}{2\pi^2} \left(\frac{\tau}{\hbar v}\right)^3 \int_0^{x_d} dx\, x^2 \frac{x\tau}{e^x - 1} = \frac{3V}{2\pi^2} \frac{\tau^4}{(\hbar v)^3} \int_0^{\theta/\tau} \frac{dx\, x^3}{e^x - 1}.$$

$$x_d = \frac{\hbar v q_d}{\tau} = \frac{\hbar v}{\tau} \left(\frac{6\pi^2 N}{V}\right)^{1/3} \equiv \frac{\theta}{\tau}$$

$$\theta = \hbar v \left(\frac{6\pi^2 N}{V}\right)^{1/3} = \text{Debye temperature.}$$

$$U = \frac{3V}{2\pi^2} \frac{\tau^4}{(\hbar v)^3} \int_0^{\theta/\tau} dx \, \frac{x^3}{e^x - 1}.$$

(1) Low-τ limit : $\frac{\theta}{\tau} \gg 1 \Rightarrow$ approximately

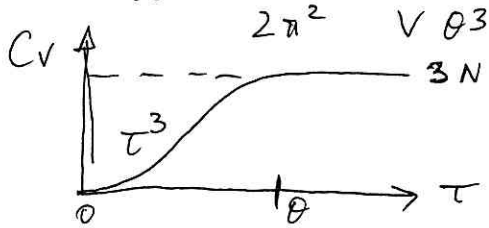
$$U = \frac{3V}{2\pi^2} \frac{\tau^4}{(\hbar v)^3} \underbrace{\int_0^{\infty} dx \, \frac{x^3}{e^x - 1}}_{\frac{\pi^4}{15}},$$

Note that $(\hbar v)^3 = \left(\frac{\theta}{\left(\frac{6\pi^2 N}{V}\right)^{1/3}} \right)^3 = \frac{V \theta^3}{6\pi^2 N}$

$$\Rightarrow U = \frac{3V}{2\pi^2} \frac{\pi^4}{15} \frac{6\pi^2 N}{V\theta^3} \tau^4 = \frac{3\pi^4 N}{5\theta^3} \tau^4$$

$$\Rightarrow C_V = \left(\frac{\partial U}{\partial \tau}\right)_V = \frac{12\pi^4 N}{5} \left(\frac{\tau}{\theta}\right)^3.$$

(2) High-τ limit : $\frac{\theta}{\tau} \ll 1 \Rightarrow$ expand e^x in denominator in small x

$$U = \frac{3V}{2\pi^2} \frac{\tau^4}{\left(\frac{V\theta^3}{6\pi^2 N}\right)} \underbrace{\int_0^{\theta/\tau} dx \, \frac{x^3}{(1+x)-1}}_{\int_0^{\theta/\tau} dx \, x^2 = \frac{1}{3}\left(\frac{\theta}{\tau}\right)^3}$$

$$\Rightarrow U = \frac{3V}{2\pi^2} \frac{6\pi^2 N}{V\theta^3} \tau^4 \frac{1}{3} \frac{\theta^3}{\tau^3} = 3N\tau \quad \text{(Dulong-Petit limit)}$$